

RAPIDSEGMENT

Transparent segments. Production-ready SQL.
No black-box trees.

Strategic Segmentation & Scorecard Engine



RapidSegment

FAST. INTELLIGENT. HIGH-IMPACT SEGMENTS.

A fast, multi-segment rule discovery and
customer segmentation engine for tabular data.



FAST
& SCALABLE



HIGH IV
SEGMENTS



AUTOMATED
RULE DISCOVERY



INTERPRETABLE
& TRANSPARENT

The problem with decision trees



Hard to load

Wide tables, mixed sources,
memory spikes



Encoding tax

One-hot / label encode kills
meaning



Unreadable

Past depth 4 no one can
explain it



Same root bias

Always starts from the same
variable

Hours vs minutes



Decision Tree path

1. Load & clean
2. Encode categoricals
3. Fit tree
4. Extract leaves manually
5. Residual loops ×4
6. Translate to SQL

4–12+ hours



RapidSegment path

1. Load with UniversalDataLoader
2. Native categoricals (no encode)
3. `extract_segments()` — one call
4. SQL filters produced
5. Scorecard JSON + weights

5–15 minutes

What stakeholders actually see

Decision Tree — not human friendly

`max_dpd_12m ≤ 15.5`

├ `util_avg ≤ 0.42`

| └ `risk_seg = Low` → Leaf A

| └ `util_avg > 0.42` → Leaf C/D

└ `max_dpd_12m > 15.5`

├ `util ≤ 0.71` → Leaf E/F

└ ... 47 more nodes ...

└ [unreadable]

RapidSegment — readable SQL

SEG 1 Lift 4.8× n=2,140

`max_dpd_12m ≥ 60`
`AND utilization ≥ 0.85`

SEG 2 Lift 3.1× n=5,820

`risk_segment = 'High'`
`AND max_dpd_12m ≥ 30`

SEG 3 Lift 2.4× n=3,450

`utilization ≥ 0.90`
`AND tenure_months < 12`

How we rank features — IV & WOE (simple)

WOE — Weight of Evidence

In plain English

How much more (or less) likely an event is in this group compared with the average.

Formula

$$WOE = \ln(\% \text{ of non-events} / \% \text{ of events})$$

Positive WOE → safer group Negative WOE → riskier group

IV — Information Value

In plain English

How useful is this whole feature at separating events from non-events?

Formula

$$IV = \sum (\%non\text{-events} - \%events) \times WOE$$

Higher IV = stronger predictor

Quick example — max_dpd_12m

Bin	% Events	% Non-events	WOE	Contribution
0 days	20%	55%	+1.01	Strong (safer)
30–59 days	25%	25%	0.00	Neutral
60+ days	55%	20%	–1.01	Strong (riskier)

IV ≈ 0.70 → Strong predictor (rule of thumb: IV > 0.30 is useful)

How it works — 6 steps

1

Rank & Bin

IV + Optimal Binning
on top features

2

Apriori

1-way → 2-way → 3-way
prune failures early

3

Grid Search

Sweep size × lift
pick global champion

4

Validate SQL

Translate bins to
ANSI SQL on residual

5

Residual

Remove matched rows
NULL-safe

6

Scorecard

Weights + deciles
to JSON

Four building blocks



UniversalDataLoader

CSV · Parquet · Excel
Arrow · BigQuery



BigQueryFeatureSelector

Screen IV in-cloud
Download only winners



StrategicSegmentBuilder

Finds high-lift rules
Outputs pure SQL



StrategicSegmentScore

Weights + deciles
JSON scorecard

Why teams choose it



No encoding

Categoricals stay categorical



Multi-way rules

1-, 2-, 3-way via Apriori



Zero overlap

Hierarchical residual guarantees it



ANSI SQL

Teams & production share one language



Cloud-first

Screen IV inside BigQuery



Full audit

Explain why a feature was selected/not selected at each step

From data to scorecard

```
from rapidsegment import StrategicSegmentBuilder, StrategicSegmentScore

builder = StrategicSegmentBuilder(target="default_flag", max_segments=5)

segments = builder.extract_segments(data)

scorer = StrategicSegmentScore(
    target_col="default_flag", primary_key="cust_id",
    segment_cols=[...], weight_type="ln_response"
)

model = scorer.calculate_and_export_weights(df, "model.json")
```

Scorecard weights

weight_type = "response"

Absolute

$$\text{weight} = \text{response_rate} \times 100$$

Best for head-hunting high-risk pockets.

weight_type = "ln_response"

Relative (log)

$$\text{weight} = \ln(\text{response_rate}) \times 100$$

Best for fair ranking across segments.

Stop explaining black boxes. Start shipping readable rules.

High-lift segments → pure SQL → calibrated scorecard.
Risk, credit, and marketing can all act with confidence.

github.com/B2BDA/RapidSegment · `pip install rapidsegment`



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